

Advanced Research Lab Report

Simulation of 3D Current-collector-designed Thick Electrode Lithium-ion Battery

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1 Introduction

The growing market and increasing demand for Lithium-ion batteries (LIBs) for applications in mobile power supplies such as portable electronics and electromobility necessitate the development of high energy density lithium-ion batteries. This goal can be achieved by two major approaches: Development of new battery chemistries with higher specific capacity, or improving the current cell structures in order to have a smaller portion of inactive mass. One way to reduce the inactive mass is to design a thick electrode (which has only one separator and two current collectors as inactive mass) instead of combining several anodes and cathodes together which needs also several separators and current collectors. The comparison between these two designs is shown in Fig. 1.

Although a thick electrode designed cell should deliver a higher capacity in comparison to a thin electrode system, there are several drawbacks to thick electrode cells which make its application challenging. Two major challenges regarding development of thick electrode cells are: 1) Fracture of the thick electrode and delamination from the 2D planar current collector (which is very likely to happen during the production process). 2) Slow electron and ion transfer in the composite electrode due to the high thickness of electrode and a highly tortuous Li^+ diffusion path (which is the phenomenon that happens during the battery application). This can lead to sluggish kinetics and poor rate performance in thick electrode battery cells. These 2 issues are also illustrated in Fig. 2.

In order to overcome the poor kinetics of thick electrode designed cells, two major solutions have been investigated. 1) Introducing hierarchical ordered pores inside the thick electrode, which results in a composite cathode microstructure with a lower tortuosity and therefore a better Li^+ transport properties. A schematic view of this approach is shown in Fig. 3. 2) Producing integrated 3D current collector/thick electrode architecture. This method increases the surface area of the current collector and the overall availability of the electrons through the electrode. The integration of the current collector and electrode

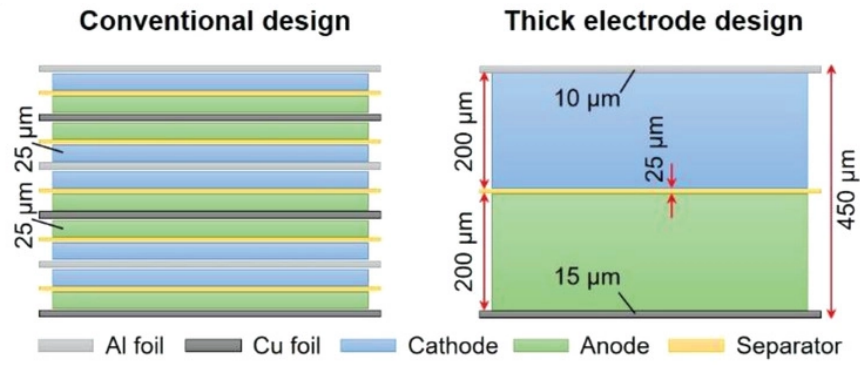


Figure 1: Ratio of components in conventional thin electrode and thick electrode cells, reproduced from [1]

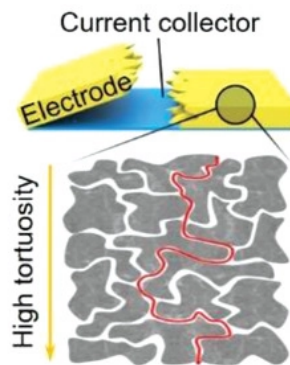


Figure 2: Drawbacks in thick electrode LIBs, reproduced from [1]

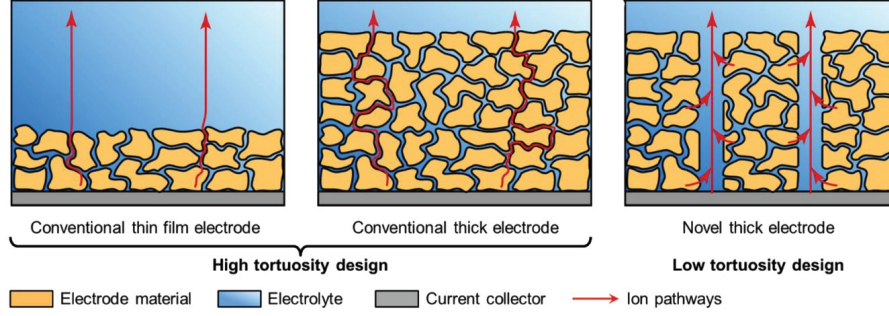


Figure 3: Low tortuosity thick electrode design, reproduced from [1]

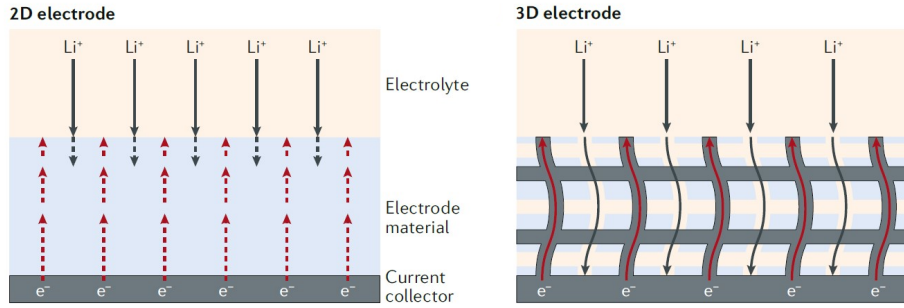


Figure 4: Integrated current collector/electrode thick electrode design, reproduced from [2]

also enhances the thick electrode's mechanical properties and current collector-electrode adhesion. An example of this design is provided in Fig. 4. There is also the possibility to combine these two methods to produce a electrode which has a low Li^+ tortuosity and also a 3D current collector.

In the experimental part of this study a woven-structured aluminum mesh with circular threads of $60\mu\text{m}$ diameter and $110\mu\text{m}$ apparatus was used as 3D current collector. This 3D Al structure was pressed to a porous free-standing cathode material (containing LiFePO_4 , carbon black and binder), in order to make an integrated 3D current collector/thick electrode cell. Schematic illustration of these 2 components of the cell and their representative microstructure are shown in Fig. 5. The goal of this study is to investigate and compare the performance of the 3D current collector cell design with conventional 2D current collector cell and also to investigate the Li^+ concentration evolution inside the electrolyte during discharge process.

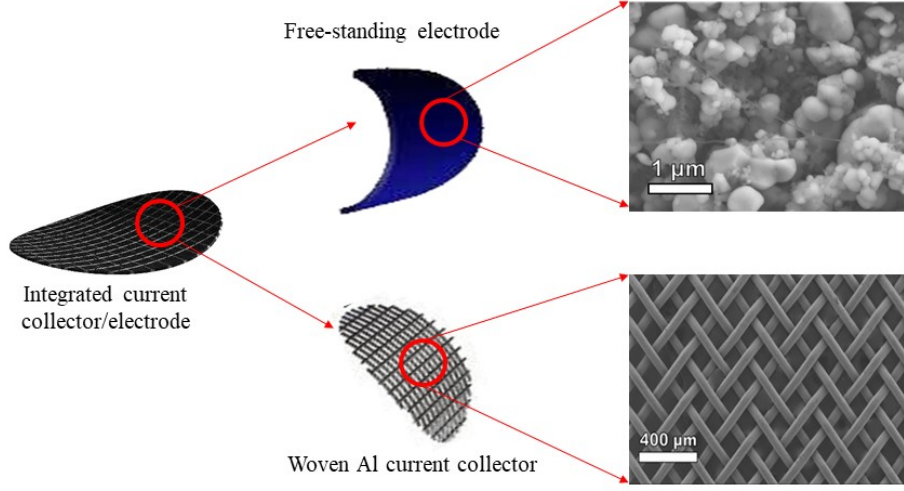


Figure 5: Integrated 3D current collector/thick electrode components

2 Simulation Method

In this work, a two-level simulation framework [3] based on the pseudo-two-dimensional (P2D) model [4] has been presented to study the cell performance of the three-dimensional (3D) LIB cell with a thick electrode. As shown in Fig. 6, the half cell consists of a separator, a homogenized composite cathode, and a 3D current collector. For each point of the composite cathode, a spherical active material particle is considered.

2.1 Models at the cell level

The degrees of freedom (DoFs) at the 3D cell level are: Li^+ ion concentration c_2 in the electrolyte, the electric potential ϕ_2 in the electrolyte phase, and the electric potential ϕ_1 for the solid phase. For the sake of computation efficiency, the mixture of active materials, the conductive carbon black as well as the binder is considered as the solid phase. Following the same notation of Doyle [4], the subscript 1 represents the quantities of the solid phase, while for the electrolyte phase, the subscript 2 is used. Governing equations at the cell level are given as follows [4, 5]:

$$\varepsilon \frac{\partial c_2}{\partial t} = D_{\text{eff}} \nabla^2 c_2 - \frac{1}{F} \mathbf{i}_2 \frac{\partial t_+^0}{\partial c_2} \nabla c_2 + (1 - t_+^0) J_{\text{eff}} \quad (1)$$

and

$$\nabla \cdot \mathbf{i}_1 + F J_{\text{eff}} = 0 \quad (2)$$

and

$$\nabla \cdot \mathbf{i}_2 - F J_{\text{eff}} = 0 \quad (3)$$

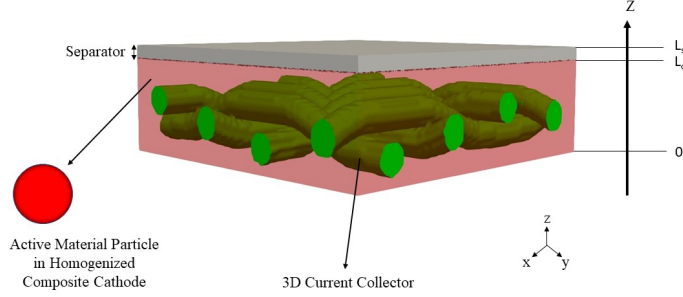


Figure 6: Schematic illustration of the half-cell

where $\mathbf{i}_1$ and $\mathbf{i}_2$ respectively denote the currents in solid and electrolyte phase, which can be stated as follows:

$$\mathbf{i}_1 = -\lambda \nabla \phi_1 \quad (4)$$

and

$$\mathbf{i}_2 = -\kappa_{\text{eff}} \left[\nabla \phi_2 - \frac{RT}{F} (1 - t_+^0) \nabla \ln c_2 \right] \quad (5)$$

On the right-hand side of Eq. (1), first term considers the Li^+ diffusion, the second part describes the Li^+ migration due to the electric field and the last term is a source term which accounts for the amount of Li^+ that takes part in the reaction with active materials. ε denotes the porosity of the composite cathode. F and R are Faraday and gas constants respectively, t_+^0 is the transference number, and J_{eff} represents the effective pore wall flux which will be explained in Section 2.3. D_{eff} and κ_{eff} respectively denote the effective diffusivity and conductivity of electrolyte in the composite cathode. In this work, the composite cathode is considered as a homogenized phase at the cell level, therefore, the homogenization for diffusivity and conductivity can be approached by using the Bruggeman relations [6]:

$$D_{\text{eff}} = \varepsilon^{1+\alpha} D_{\text{intrinsic}} \quad (6)$$

$$\kappa_{\text{eff}} = \varepsilon^{1+\alpha} \kappa_{\text{intrinsic}} \quad (7)$$

where α is the Bruggeman coefficient. It should be noted that, for the separator part in the cell level, the pore wall flux term in Eqs. (1) to (3) is zero and also in Eqs. (1) and (3) the effective diffusivity and conductivity equal the intrinsic values, namely $\varepsilon = 1$. The related boundary conditions for the cell level's problem are presented in Table 1, where a constant current I_{app} is applied for the galvanostatic discharge process.

Table 1: Boundary conditions of the cell level

Condition	Boundary
$-D_{\text{eff}} \nabla c_2 \cdot \mathbf{n} = \frac{I_{\text{app}}(1-t_+^0)}{F}$	$Z = L_c + L_s$
$-D_{\text{eff}} \nabla c_2 \cdot \mathbf{n} = 0$	$Z = 0$
$i_1 \cdot \mathbf{n} = -\lambda \nabla \phi_1 \cdot \mathbf{n} = I_{\text{app}}$	surface of 3D current collector
$i_1 \cdot \mathbf{n} = -\lambda \nabla \phi_1 \cdot \mathbf{n} = 0$	$Z = L_c + L_s$
$i_2 \cdot \mathbf{n} = -\kappa_{\text{eff}} [\nabla \phi_2 - \frac{RT}{F}(1-t_+^0) \nabla \ln c_2] \cdot \mathbf{n} = I_{\text{app}}$	$Z = L_c + L_s$
$i_2 \cdot \mathbf{n} = -\kappa_{\text{eff}} [\nabla \phi_2 - \frac{RT}{F}(1-t_+^0) \nabla \ln c_2] \cdot \mathbf{n} = 0$	surface of 3D current collector
$\phi_1 = 0$	$Z = L_c + L_s$

2.2 Particle Level

In this level, the transport of lithium inside the active material particle is considered. The governing equation at the particle level is a diffusion equation based on Fick's second law and no feature of phase separation inside the LiFePO_4 particles is considered. The concentration of Li, namely c_1 , is the only interested DoF at this level. Therefore, the mass conservation of lithium inside the particle can be expressed as follows:

$$\frac{\partial c_1}{\partial t} = D_1 \nabla^2 c_1 \quad (8)$$

The pore wall flux J_s calculated from the Butler-Volmer equation is applied to the surface of particle as the boundary condition as follows:

$$-D_1 \nabla c_1 \cdot \mathbf{n} = J_s \quad (9)$$

where $\mathbf{n}$ denotes the normal vector of the outward surface of the particle. Similar to the work of Bai et al. [3], the pore wall flux is treated as a dependent quantity in this work, and can be calculated as follows:

$$J_s = \frac{i_0}{F} [c_{1,\text{surface}} \exp(\frac{\alpha_a F \eta}{RT}) - (1 - c_{1,\text{surface}}) \exp(-\frac{\alpha_c F \eta}{RT})] \quad (10)$$

where $c_{1,\text{surface}}$ is the concentration of Li at the surface of the particle. α_a and α_c are the anodic and cathodic charge transfer coefficients, in this work they are both considered as 0.5, namely $\alpha_a = \alpha_c = 0.5$. The exchange current density i_0 , and the overpotential η can be given out as follows:

$$i_0 = F k_2 (c_{2,\text{max}} - c_2)^{\alpha_c} (c_2)^{\alpha_a} = F k_2 \sqrt{(c_{2,\text{max}} - c_2) c_2} \quad (11)$$

and

$$\eta = \phi_1 - \phi_2 - U_{oc} \quad (12)$$

where k_2 denotes the reaction rate constant. U_{oc} is the open circuit potential which is a function of $c_{1,surface}$. In this work, the U_{oc} for LiFePO_4 material, which is adopted from experimental results [7], can be stated as follows:

$$U_{oc} = 3.114559 + 4.438792 \tan^{-1}\left(-71.7352 \frac{c_{1,surface}}{c_{1,max}} + 70.85337\right) - 4.240252 \tan^{-1}\left(-68.5605 \frac{c_{1,surface}}{c_{1,max}} + 67.730082\right) \quad (13)$$

2.3 Bridging Level

The pore wall flux calculated using equation Eq. (10), takes values of c_2 , ϕ_1 , and ϕ_2 from the cell level, as well as $c_{1,surface}$ from the particle level as input. Therefore, the calculated pore wall flux is used in both levels, in particle level it serves as the boundary condition, while in the cell level it plays as the source term to account for the chemical reaction. In the cell level's equations, the effective pore wall flux, J_{eff} is used, which can be obtained from the following relation:

$$J_{eff} = aJ_s \quad (14)$$

with a being the specific surface area and equals:

$$a = \frac{S}{V}(1 - \varepsilon) \quad (15)$$

where S and V respectively denote the surface area and the volume of the particle. The dependence of the pore wall flux on the variables of both levels and its usage in both levels models make the bridging possible.

The software GeoDict® is used to produce 3D cell level geometry and Multiphysics Object-oriented Simulation Environment (MOOSE) [8], and its multi-scale capability [9] is used to solve the coupled partial differential equations (PDEs) by using the finite element method (FEM) in two levels.

3 Results and Discussion

In this work two different half cell designs are considered. One is a 3D domain with the size of $336\mu\text{m} \times 336\mu\text{m} \times 228\mu\text{m}$, which has a 3D woven-structured cathode current collector. While the other one is a 3D cell with the same volume as the previous one but a 2D planar conventional cathode current collector. Each of these half cells consists of a separator, electrolyte and a composite cathode, composite cathode consists of spherical particles with $1\mu\text{m}$ radius (also illustrated schematically in Fig. 6). The following results are based on considering LiPF_6 as the electrolyte and LiFePO_4 as the cathode material. The parameters we used in the simulations are listed in Table 2, it is worth mentioning that the model is general and other material systems can also be considered.

Table 2: Parameters used in simulation [4, 5, 10, 11]

Symbol	Value	Unit
D_1	1.18×10^{-18}	$\text{m}^2 \text{s}^{-1}$
λ	0.5	S m^{-1}
D_2	7.5×10^{-11}	$\text{m}^2 \text{s}^{-1}$
κ	1.5×10^{-1}	S m^{-1}
ε	0.2	-
R	8.314	$\text{J mol}^{-1} \text{K}^{-1}$
T	298.15	K
F	96485	C mol^{-1}
t_+^0	$0.0107907 + 1.48837 \times 10^{-4} c_2$	-
α (Bruggeman coefficient)	0.5	-
i_{app} (1 C-rate)	82.18	A m^{-2}

3.1 Cell performance comparison of two cell designs

Fig. 7 shows the cell performance of the two mentioned cell designs at different C-rates under discharge process. As it can be seen in this figure, at all the considered C-rates, 3D current collector designed cell delivers a higher capacity than the conventional current collector designed cell (almost 1.36 times larger in 1C which is in accordance to experimental results). This higher capacity can be attributed to the higher surface area of the cathode current collector and its 3-dimensional distribution through the composite cathode in 3D current collector cell. Therefore the reaction kinetics is enhanced and the amount of cathode active material which takes part in the reaction and becomes lithiated is increased. Another consequence is the reduction in the cell resistance, which can also be seen in the diagrams of Fig. 7.

Higher capacity and smaller polarization for 3D current collector designed cell in comparison to conventional current collector cell is also observed during charging process which is depicted in Fig. 8. It is worth mentioning that, the difference in the covered range of SOC in these two plots is the main reason for the different elapse time in Figs. 7 and 8.

3.2 Comparison of lithium ion distribution in electrolyte

As it can be seen in Fig. 9, there is a larger amount of Li^+ diffuse to the further parts of the electrolyte in 3D current collector designed cell than the conventional current collector cell. Since no difference in tortuosity and diffusivity

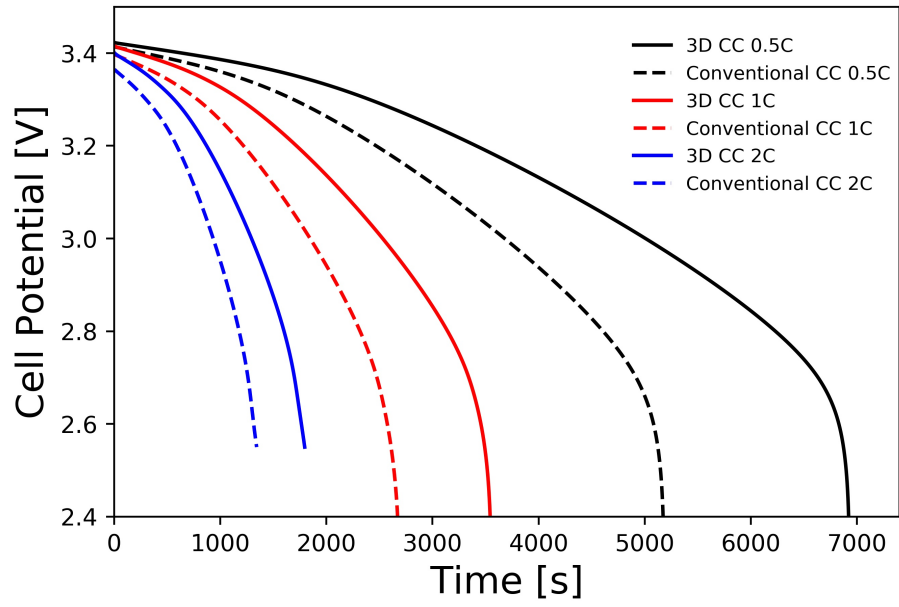


Figure 7: Evolution of cell potential at different C-rates for 3D current collector (3D CC) and conventional current collector (conventional CC) cell designs during discharge for SOC 0.46-1

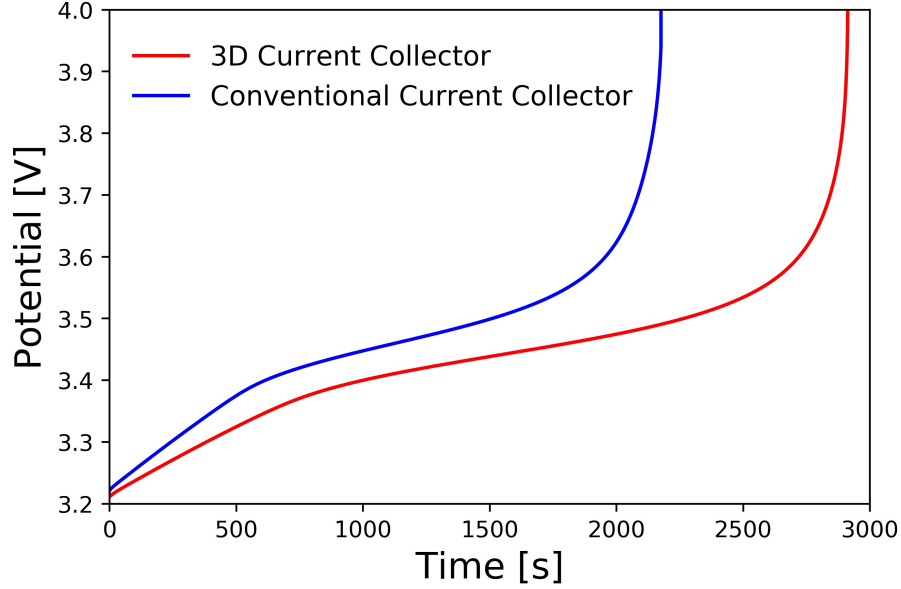


Figure 8: Cell potential of 3D current collector and conventional current collector designed cells during charging process with 1C for SOC range of 0.97-0.03

of Li^+ in electrolyte are considered, the reason for this further transport is the faster kinetics of the 3D current collector cell, wherein Li^+ is consumed faster in the discharge reaction which contributes to a higher gradient for Li^+ diffusion.

It is also confirmed from Fig. 7 that, in 3D current collector designed cell, parts of the electrolyte which are far from the separator experience lower Li^+ concentration in comparison to the conventional current collector cell. This can also be attributed to the faster kinetics of the discharge reaction in 3D current collector cell. At higher currents, when the movement of Li^+ ions from the separator cannot keep up with the consumption of Li^+ in the discharge reaction, it could be a problem for the cell. In order to solve this problem, the electrolyte in the 3D current collector cell is necessary to have a higher effective Li^+ diffusivity, which is achievable by means of reducing the tortuosity. Moreover, this will also reduce the current inhomogeneity in Li^+ distribution observed in 3D current collector cell.

4 Conclusion

In this work we introduced a two-level framework to simulate the charge and discharge process of 3D current collector designed thick electrode LIB and also to compare it with conventional 2D planar current collector thick electrode LIB. Results show that 3D current collector cell delivers a higher capacity and it has

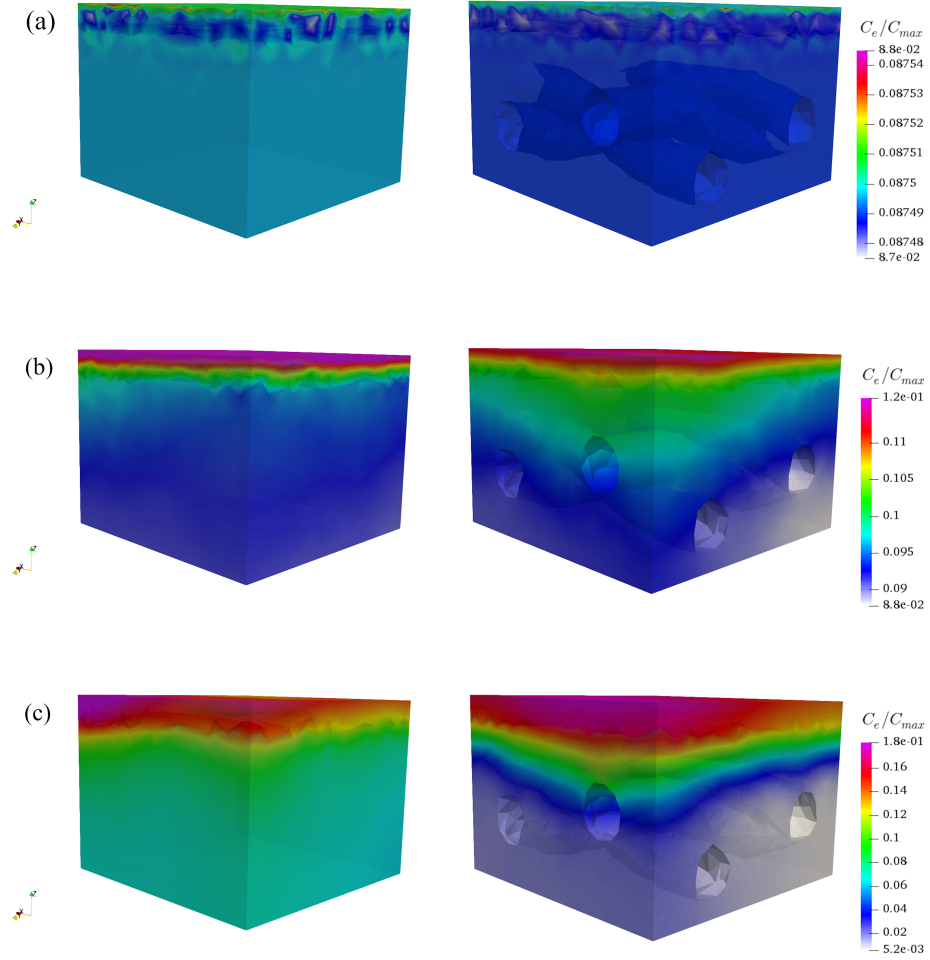


Figure 9: Lithium ion concentration profiles in electrolyte in 3D current collector designed (right) and conventional designed (left) cells at (a) beginning of the discharge process, (b) at the cell potential of 3 V, and (c) at the end of the discharge process

a deeper but a less homogeneous Li^+ diffusion through the cell.

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